MODULE 01

Lumbar Spine MRI: Disc, Stenosis & Endplate Grading

Pfirrmann · Disc Nomenclature · Canal & Foraminal Stenosis · Modic · Meyerding

1.0 AMA PRA Cat 1 | **Spine** | **CME Article Review**

LEARNING OBJECTIVES

1. Apply Pfirrmann grading (I–V) to lumbar disc degeneration on sagittal T2 MRI.
2. Classify disc herniations using ASNR nomenclature and axial zone location.
3. Grade central canal stenosis using Lurie (rule of thirds) and Schizas (A–D).
4. Grade neural foraminal stenosis using the Lee classification (0–3) on sagittal T1.
5. Identify Modic endplate types 1–3 and their BVN ablation implications.
6. Apply Meyerding spondylolisthesis grading (I–V) and management thresholds.

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1. Pfirrmann Disc Degeneration — Sagittal T2

Five-grade system on sagittal T2 MRI assessing signal, structure, NP/AF distinction, and disc height. Grades III–V indicate degeneration of increasing severity.

Grade	Signal	NP/AF Distinction	Height	Clinical
I — Normal	Bright white homogeneous	Clear	Normal	No degeneration
II	Inhomogeneous ± gray bands	Clear	Normal	Early change
III	Inhomogeneous gray	Unclear	Normal–slightly ↓	Equivocal
IV	Inhomogeneous gray-black	LOST	Normal–moderately ↓	Degenerate
V — Collapsed	Homogeneous black	LOST	Collapsed	End-stage

KEY POINT

Grades I–II = normal. Grade III = equivocal. Grades IV–V = significant degeneration. Always correlate Pfirrmann with Modic changes and stenosis at the same level for full clinical picture.

References

• Pfirrmann CW et al. Spine 2001;26(17):1873–1878.

2. Disc Herniation Nomenclature — ASNR/AANS 2014

Herniation Morphology

Term	Morphology	Key Feature
Broad-based bulge	>180° circumferential ≥2 mm extension	NOT a herniation; no focal component
Protrusion	Base WIDER than dome	Focal; in continuity; stable morphology
Extrusion	Dome WIDER than neck	In continuity; "pseudoaneurysm" shape
Sequestration	Free fragment; lost continuity	May migrate; may enhance; any direction

Zone Location (Axial) — Root Compressed

Zone	Location	Root Compressed	Risk
Central	Midline canal	Bilateral roots / cauda equina	CES risk

Zone	Location	Root Compressed	Risk
Subarticular / Paracentral	Lateral recess	TRAVERSING root (level below)	L4-5→L5
Foraminal	Within neural foramen	EXITING root (same level)	L4-5→L4
Extraforaminal	Lateral to foramen	EXITING root	Lateral approach

CRITICAL

Paracentral herniations compress the TRAVERSING root (one level below). Foraminal herniations compress the EXITING root (same level). Failure to distinguish leads to wrong-level surgery.

3. Central Canal Stenosis — Lurie & Schizas

Lurie — Rule of Thirds (Axial T2)

Grade	Thecal Sac Obliteration	Roots	Implication
0 — Normal	None	Unrestricted halo	Asymptomatic
1 — Mild	< 1/3	All visible	Intermittent claudication possible
2 — Moderate	1/3–2/3; clumping	Reduced	Neurogenic claudication
3 — Severe	> 2/3 obliterated	Massed	Surgical candidate

Schizas — Morphological (Axial T2)

Grade	CSF	Roots	Severity
A	Homogeneous; present	Visible	Normal
B	Inhomogeneous; present	Visible	Mild
C	Absent	Still distinguishable	Moderate–severe
D	Absent	Indistinguishable (solid block)	Extreme

4. Neural Foraminal Stenosis — Lee (Sagittal T1)

Grade	Perineural Fat	Nerve Root	Clinical
0 — Normal	Clear; surrounds root	Undeformed	No stenosis
1 — Mild	Partially obliterated	Not deformed	Possible radiculopathy
2 — Moderate	Completely obliterated	Deformed	Radiculopathy correlates

Grade	Perineural Fat	Nerve Root	Clinical
3 — Severe	Completely obliterated	Invisible	Strong radiculopathy

5. Modic Endplate Changes & Meyerding Spondylolisthesis

Modic Classification (Sagittal T1 + STIR)

Type	T1	STIR	Pathology	BVN Target
1 — Edema	↓ Hypointense	↑ Hyperintense	Active inflammation; vascular fibrous	✓ YES (strongest)
2 — Fatty	↑ Hyperintense	↓ Suppressed	Fatty marrow replacement	✓ YES
3 — Sclerosis	↓ Hypointense	↓ Hypointense	Dense bone; subchondral sclerosis	✗ NO

Meyerding Spondylolisthesis Grading

Grade	Slip (%)	Management
I	0–25	Conservative first
II	26–50	Conservative; surgical if refractory
III	51–75	Instrumented fusion usually required
IV	76–100	Instrumented fusion; complex
V — Spondyloptosis	>100	Complete reconstruction

CRITICAL

Distinguish Modic 1 from discitis: discitis shows disc T2 signal, height loss, and paraspinal/epidural extension. Modic 1 has preserved height and no soft tissue spread. Both can show endplate enhancement.

References

• Modic MT et al. Radiology 1988;166(1):193–199.
• Fischgrund JS et al. Spine 2018;43(17):1176–1183.